

# BanglaWeatherBench: Monsoon-Stratified Evaluation of Zero-Shot Time-Series Foundation Models on Station Observations

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September 18, 2026

## Abstract

Time-series foundation models are presented as zero-shot forecasters for arbitrary series, but the evidence is drawn overwhelmingly from mid-latitude, data-rich and frequently reanalysis-derived sources. We introduce BanglaWeatherBench, a benchmark of daily weather forecasting on tropical-monsoon station observations: 16 tasks and 170,571 rolling-origin windows over 35 Bangladesh Meteorological Department stations, 10 co-located GHCN-Daily stations, 32 European GHCN-Daily stations as a temperate reference, and NASA POWER reanalysis at all 77 station coordinates, with an independent CHIRPS dekadal rainfall task. We score 8 zero-shot foundation models against 10 trained and statistical baselines, resampling by forecast origin with Diebold–Mariano tests and Holm correction across 48 comparisons. Foundation models are the strongest family, yet the best one separates from the best baseline in only 14 of 48 comparisons, loses 5, and is indistinguishable in 29; on station observations it wins 3 of 24, all at a one-day lead, against 11 of 24 on reanalysis at the same coordinates. No model beats a day-of-year climatology on rainfall at the 35 stations beyond a one-day lead, or on dekadal rainfall at any horizon. Reanalysis looks more predictable than the stations it represents in 78 of 108 comparisons. At a 30-day lead, 4 of 8 models lose significantly more temperature skill in Bangladesh than a gap-matched trained control, with roughly 60% of that penalty attributable to sparser station context rather than to climate. Skill also reverses by monsoon regime, and the full leaderboard reproduces in 4.16 GPU-hours on a free GPU.

## 1 Introduction

Time-series foundation models promise a particular kind of generality: pretrain once on a large heterogeneous corpus, then forecast an unseen series with no fitting, no feature engineering and no local expertise. If that promise holds where forecasting capacity is scarcest, it matters. The places with the thinnest station networks, the fewest trained forecasters and the least compute are often the places where a week of warning carries the largest consequences, and a model that runs zero-shot on a free GPU is precisely the technology they lack.

Whether it holds there is an empirical question current benchmarks do not answer, because the evidence base is skewed in two ways at once. Geographically, general time-series suites treat weather as one domain among many and draw it overwhelmingly from mid-latitude sources [2, 13, 18, 29]; across seven widely used benchmark papers, the words “monsoon” and “Bangladesh” do not appear. Representationally, the weather benchmarks that *are* globally comprehensive score against reanalysis [27, 30], a physically consistent reconstruction that is smoother than the station record it interpolates, and smoothest exactly where stations are sparsest. Observation-based evaluation exists for numerical and AI weather-prediction models [14, 16], but the corresponding question for zero-shot time-series models has not been asked.

Bangladesh makes the question concrete. Its weather is governed by the South Asian summer monsoon, whose defining feature, the alternation of active and break spells inside the rainy season, has no temperate analogue. In the 35-station record we build on, median July rainfall reaches 457 mm per station-month against 0 mm in January, 67.6 % of station-days are dry, and the wet-day 99th percentile is 143 mm; the 2016–2023 test period alone contains 432 cyclone station-days across 14 storms. The record that would support decisions here is thin: over the same period the ten Bangladeshi stations in GHCN-Daily are about 65 % complete, with 2021 almost entirely absent, against 99 % at the European stations we use as a reference. That asymmetry is not only a property of some pretraining corpus: it sits inside the model’s input window at forecast time.

We introduce **BanglaWeatherBench**, a benchmark designed so that its headline number is not the only thing it produces. It defines 16 daily forecasting tasks over 170,571 rolling-origin windows across four tracks: 35 Bangladesh Meteorological Department stations, 10 co-located GHCN-Daily stations, 32 European GHCN-Daily stations as a temperate reference, and NASA POWER reanalysis at all 77 station coordinates, plus an independent CHIRPS dekadal rainfall task over 64 districts. Three controls answer the confounds that would otherwise make a tropical result unreadable: a same-network temperate reference isolating climatic regime, a reanalysis track at identical coordinates isolating the effect of the scoring target, and a gap-matched context control, applied to the trained baselines as well as the foundation models, isolating data availability from climate. Scoring is stratified by monsoon phase and by active/break spell.

Two commitments shape what we are willing to claim. First, significance: a 7-day stride with a 30-day horizon produces heavily overlapping windows, so we resample at the forecast origin and report Diebold–Mariano tests with HAC variances, moving-block bootstrap intervals, Friedman–Nemenyi ranks and Holm correction across 48 comparisons. Many margins that look decisive as point estimates do not survive. Second, leakage: we audit each model’s pretraining corpus against its technical report. Leakage-free zero-shot evaluation is not itself novel [28]; here it is a precondition for reading a geographic contrast as a geographic effect.

Evaluating 8 zero-shot foundation models against 10 trained and statistical baselines yields a mixed result that we think is more useful than either a triumphal or a dismissive one.

1. **Foundation models lead, but most of the lead is not separable.** The best model beats the best baseline in 14 of 48 Holm-corrected comparisons, loses 5 and is indistinguishable in 29; on station observations it wins 3 of 24, all at a one-day lead, against 11 of 24 on reanalysis. The six leading models and PatchTST sit within one critical difference.
2. **Nothing beats climatology on tropical rainfall** beyond a one-day lead at the BMD stations, or at any horizon on dekadal rainfall, a concrete boundary on the “forecast anything” claim.
3. **Reanalysis flatters models** relative to the stations it represents, in 78 of 108 comparisons.
4. **The tropical temperature deficit is narrow and mostly about data.** At a 30-day lead 4 of 8 models lose significantly more skill than a like-for-like control; about 60 % of the raw penalty is explained by sparser context, and doubling the context does not remove the remainder.
5. **Skill reverses by regime:** strong in break spells, significantly worse than climatology in active spells, in 61 of 72 tests.

We state finding 4 carefully because it is easy to misreport. The evidence supports a specific claim: for temperature at long horizons, half the roster loses significantly more skill on Bangladeshi stations than trained baselines do, most of it traceable to sparser context. It does

not support a blanket claim that foundation models fail in the Global South. Stated correctly it is also the more actionable result: much of the deficit is a data-availability problem that observation programmes and gap-tolerant architectures can address.

**Contributions.** (i) A released, leakage-tested benchmark for a tropical-monsoon climate, with splits, strata, a datasheet and per-source licensing. (ii) Controls rather than a leaderboard alone: a same-network temperate reference, a reanalysis track at identical coordinates, and a gap-matched control applied to models and baselines alike. (iii) A significance protocol matched to overlapping rolling-origin panels, applied uniformly. (iv) To our knowledge the first evaluation of current zero-shot time-series foundation models on South Asian station observations including precipitation, stratified by monsoon phase, with a pretraining-overlap audit. (v) Reproducibility at the compute scale of the places the benchmark is about: 4.16 GPU-hours on a free Tesla T4.

## 2 Related work

**General time-series benchmarks.** Monash [13], GIFT-Eval [2], TFB [29] and TSFM-Bench [18] evaluate across many domains and report aggregate ranks. Weather appears as one domain among dozens, drawn mostly from mid-latitude sources, and none stratifies by climatic regime or geography. TIME [28] adds a strict leakage-free protocol over 50 fresh datasets and 12 foundation models, and likewise applies no geographic or climatic strata. Recent work has begun to document that aggregate metrics conceal regime-dependent failure [32] and that benchmark construction itself is a source of leakage [25]; our design follows both warnings.

**Weather benchmarks.** WeatherBench 2 [30] and ChaosBench [27] score against reanalysis; ExtremeWeatherBench [23] scores against event catalogues; WeatherReal [16] and WEATHER-5K [15] score against station observations. WEATHER-5K is the closest in spirit (thousands of global stations, hourly), but carries neither precipitation nor humidity, which are the variables at issue here, and reports global aggregates rather than climatic strata. None of these evaluates time-series foundation models on a tropical station network.

**Geographic disparity in AI forecasting.** SAFE [22] stratifies AI weather-prediction models by country, subregion and income group, finding disparities across every attribute, but works on ERA5 at  $1.5^\circ$  resolution for upper-air fields only. Mozaffari et al. [26] argue qualitatively that AI risks amplifying a North–South divide without supplying measurements. Rowell Jr. and Kupssinskü [31] stratify GraphCast by climatic sub-region over Brazil against operational IFS analysis for  $T_{850}$ ,  $Q_{850}$  and  $Z_{500}$ , excluding precipitation. Ours is the corresponding measurement for time-series foundation models, on station observations, including rainfall.

**South Asian monsoon and AI models.** Gupta et al. [14] assess seven global AI weather-prediction models against South Asian observations during the monsoon, reporting errors 15–45% larger than against reanalysis, the precedent for our observation-versus-reanalysis contrast, transferred here to a different model class. Luitel et al. [21] evaluate GraphCast over the Indian summer monsoon against ERA5 and IMERG, and Aitken et al. [1] build decision-oriented probabilistic onset forecasts for Indian farmers from AI weather models. All three concern gridded NWP-style emulators and onset or seasonal-mean skill rather than zero-shot station forecasting.

**Precipitation-specific evaluation.** A recent benchmark evaluates 17 supervised architectures for 0–6 hour GNSS-based precipitation nowcasting [34]. The task, horizon and model

class all differ from ours, but it is the closest existing precipitation-focused time-series benchmark.

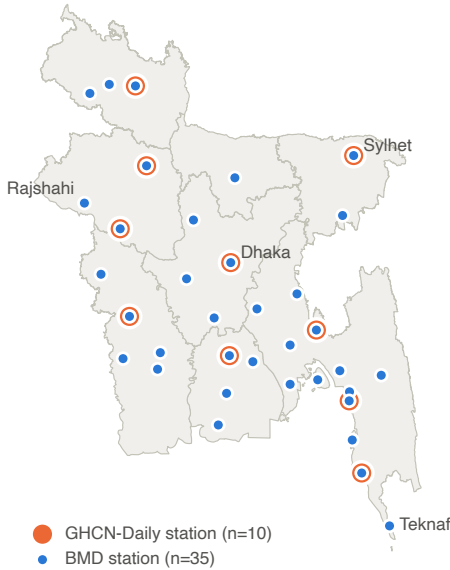
**Bangladesh forecasting.** The applied literature is substantial but uses per-station supervised models and releases no shared tasks: Joy et al. [17] is representative, training attention-enhanced LSTMs for temperature and rainfall. No published work evaluates pretrained foundation models on Bangladeshi station data, and no Bangladesh weather benchmark with fixed splits and a leaderboard exists.

**Position.** The intersection is, to our knowledge, unoccupied: current time-series foundation models, on tropical station observations including precipitation, stratified by monsoon regime, with a matched temperate control and a significance protocol. We claim novelty only at that intersection, and cite the neighbours above for each component individually.

### 3 BanglaWeatherBench

The benchmark is built around one observational target and three controls, each answering a confound that would otherwise make a tropical result unreadable.

**a Bangladesh observation stations**



**b GHCN-Daily tracks: same WMO network, two climates**

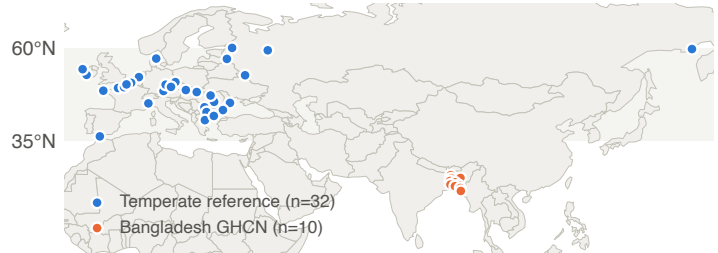


Figure 1: The three station networks. Left: 35 BMD stations across Bangladesh (the forecast target), with the 10 co-located GHCN-Daily Bangladesh stations marked. Right: the 32 mid-latitude European GHCN-Daily stations that form the temperate reference. Both station sets are drawn from instrument networks reporting to the same WMO standards, which is what allows climatic regime to be varied while the observing protocol is held fixed. Administrative boundaries for Bangladesh are the OCHA/HDX Common Operational Datasets (CC BY-IGO); world boundaries are Natural Earth (public domain).

#### 3.1 Data tracks

**BMD stations (target).** Daily observations from 35 Bangladesh Meteorological Department stations, 1961–2023, covering rainfall (mm), temperature ( $^{\circ}\text{C}$ ), relative humidity (%) and bright

sunshine (hours), released under CC-BY-4.0 [35]. The published record was already imputed by its authors using forward/backward filling, interpolation and manual edits, so we detect and flag suspect values rather than trusting the file: constant runs, perfectly linear runs, and values lying off the recording grid. Over the test period, flagged shares are 0.08 % (rainfall), 1.26 % (temperature), 1.75 % (sunshine) and 4.44 % (humidity); the worst single station-variable is 14.1 %, and only 11 of 140 station-variables exceed 5 %. A window is scored only if its target is complete *and* unflagged, so flagged values never enter a score.

**GHCN-Daily Bangladesh (cross-check).** Ten stations from the WMO-derived network [24], co-located with BMD sites, giving an independently curated view of the same weather. Over 2016–2023 these are about 65 % complete, with 2021 almost entirely absent. GHCN labels a rainfall total by the day the 24-hour accumulation begins and BMD by the day it ends; we correct the one-day offset only for cross-source comparison, never for per-track forecasting.

**GHCN-Daily temperate (reference).** Thirty-two mid-latitude European stations from the same network, selected for record length and completeness (about 99 % over the test period). Holding the instrument network, variables, protocol and scoring fixed isolates climatic regime from everything else. All 32 are European; none is in the United States, which matters for the leakage audit (§4).

**NASA POWER (reanalysis).** Reanalysis sampled at all 77 station coordinates, so that observation and reanalysis can be compared at identical places and dates rather than across different domains.

**CHIRPS dekadal (independent source).** Ten-day rainfall totals for 64 districts, 1981–2026, from a satellite-gauge product. Two administrative units are split across polygons sharing one identifier; we merge them with pixel-count weights rather than dropping either.

## 3.2 Tasks

Sixteen daily tasks are defined as the cross of tracks and variables: four BMD variables, two GHCN-Bangladesh (rainfall, mean temperature), two GHCN-temperate, and the four NASA POWER analogues at each of the three station sets. A seventeenth task covers CHIRPS dekadal rainfall at a six-dekad horizon. Tasks are evaluated separately; we never pool across tracks, because scaled errors are not comparable between a smooth reanalysis grid and a sparse station record.

## 3.3 Evaluation strata

Scoring is stratified by physically defined, externally validated regimes rather than by empirical clusters. Monsoon phases (pre-monsoon, onset, peak, withdrawal, dry) follow published Bangladeshi climatology, and active/break spells follow Ferdoushi et al. [9]: runs of at least three days in July–August where daily rainfall on the all-station mean series sits above or below half a standard deviation of the daily climatological cycle. Our labels reproduce that paper’s counts to within a few percent. Strata are derived from the training period only and are used for scoring alone; they are never model inputs.

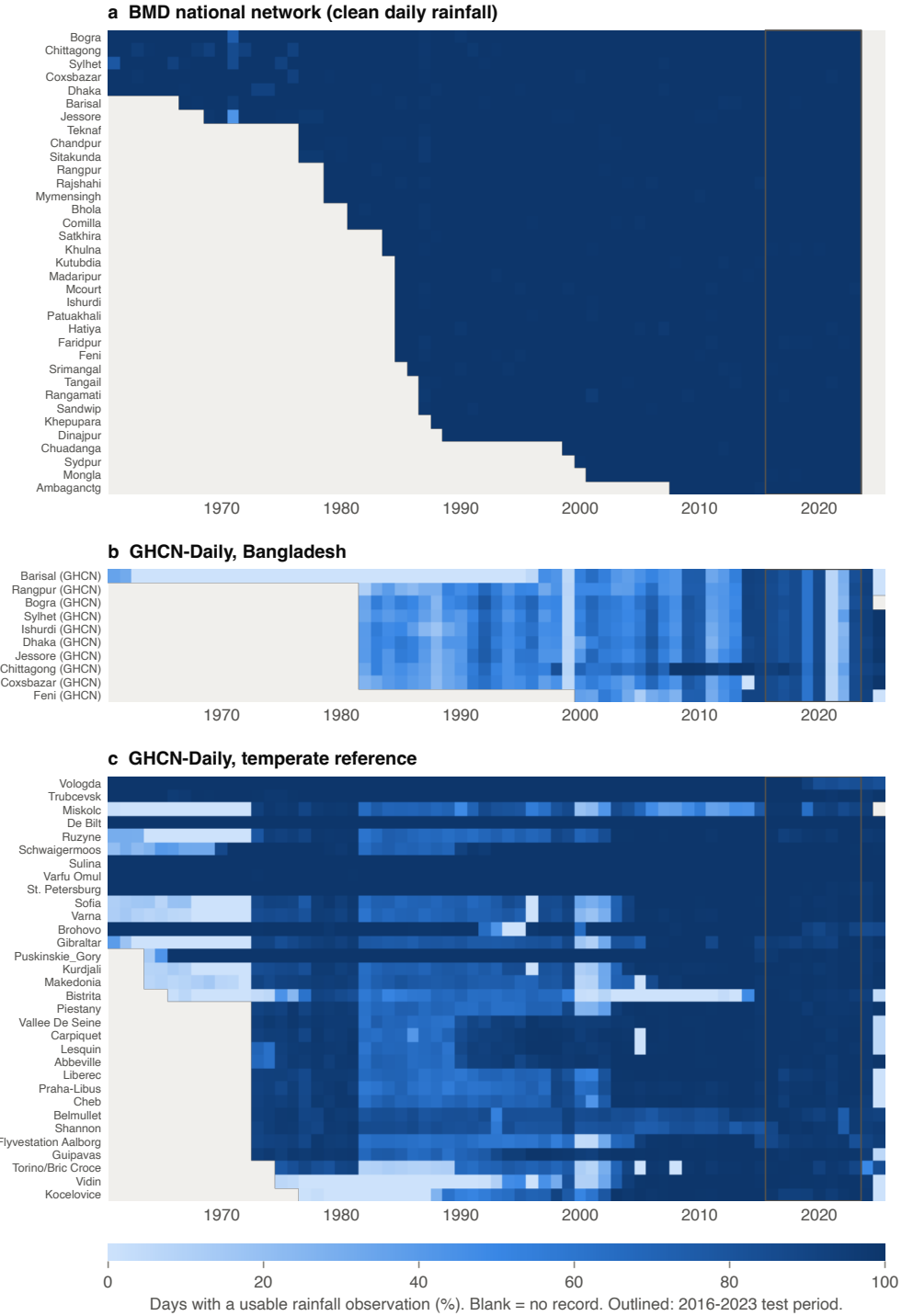


Figure 2: Usable observations per station-year, for the three station tracks. The 2016–2023 test period is outlined. The BMD target record is dense; the GHCN-Daily Bangladesh track is roughly 65 % complete over the test period with 2021 almost entirely absent, against 99 % for the temperate reference. This gap is a property of the observing system rather than of the climate, and it is what the gap-matched control in §4 exists to neutralise.

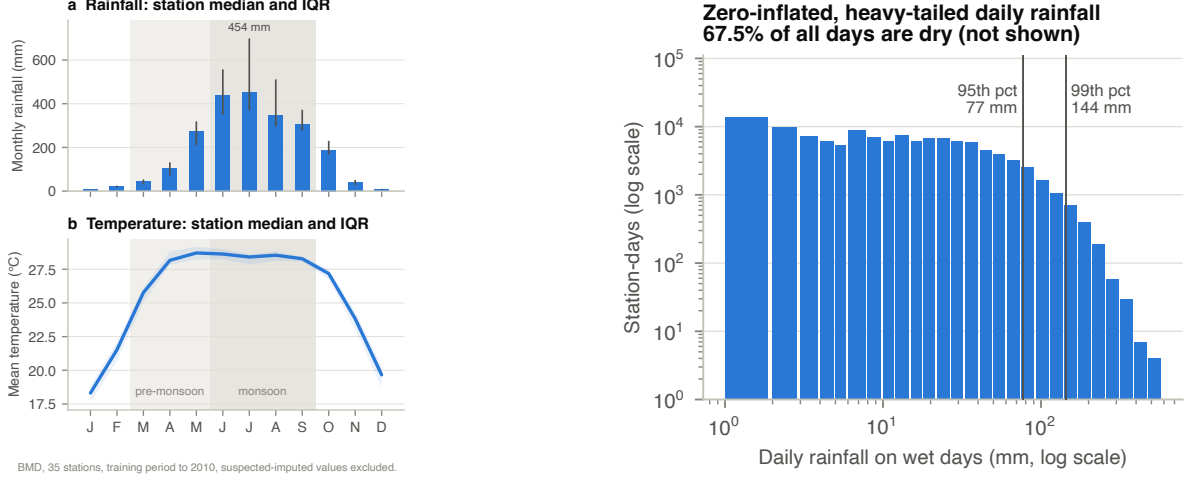


Figure 3: Two properties of the target record that shape the results. Left: the annual cycle of rainfall and temperature across the 35 BMD stations over the training period, showing the monsoon window the phase strata partition. Right: daily rainfall on wet days, log–log. Two thirds of station-days are dry and the wet tail runs past 500 mm, so a rainfall forecast is mostly a forecast of occurrence; this is the distribution against which climatology proves hard to beat (§5).

## 4 Evaluation protocol

### 4.1 Splits and windows

Temporal splits are fixed for every track: training up to 2010-12-31, validation 2011–2015, and a 2016–2023 test period common to all tracks. Anything estimated from data (climatologies, percentile thresholds, the spell reference cycle, and model fitting) uses the training split only.

Evaluation is rolling-origin with a 7-day stride and horizons of 1, 3, 7, 14 and 30 days. A window is scored when its 30-day target lies entirely inside the test split, is complete and unflagged, and at least 365 days of history precede the origin. Context is deliberately *not* required to be complete: a 90%-complete-context rule leaves zero admissible GHCN-Bangladesh rainfall windows, which would silently delete the sparsest and most representative part of the problem. Models therefore receive contexts containing real gaps, and each window records its context completeness so that skill can be stratified by it. This yields 170,571 daily windows and 18,112 dekadal windows.

### 4.2 Metrics

The headline metric is the continuous ranked probability score, a strictly proper scoring rule [12], computed here from nine predictive quantiles as twice the mean pinball loss. Point errors use each model’s predictive *median* and squared errors its mean; mixing the two is not neutral on zero-inflated rainfall, where scoring climatology on its mean rather than its median changed its MASE at a 30-day lead from 0.58 to 0.82 and made a quantile-loss network appear 25 % better than it is. Interval calibration is reported as coverage of the central 80 % interval.

Because scaled errors differ systematically between tracks, all cross-track comparisons use skill relative to the *same track’s* day-of-year climatology, estimated on the training split. A score of zero is climatology.

### 4.3 Significance

Neighbouring origins overlap heavily, so treating windows as independent samples would overstate confidence substantially; none of the reference forecasting benchmarks reports significance tests at all. We therefore make the forecast origin the unit of resampling, averaging stations within an origin so that cross-sectional correlation is absorbed rather than counted as extra samples.

Pairwise comparisons use Diebold–Mariano on the paired loss differential, with a Newey–West HAC variance at lag  $h - 1$  and the Harvey–Leybourne–Newbold small-sample correction, referred to a  $t$  distribution. Interval estimates come from a moving-block bootstrap over origins (block 8 origins, 2,000 resamples); paired series (a model, its climatology reference and the control baseline) are resampled with the same blocks, so a difference-in-differences interval carries the control’s uncertainty rather than treating it as fixed. Rankings use Friedman with the Nemenyi critical difference from the studentized range. All 48 task-and-lead comparisons are Holm–Bonferroni corrected, and we report the corrected values.

### 4.4 Controls

**Gap-matched contexts.** Bangladeshi contexts are 82–86% observed over the year preceding an origin, against 99–100% for the temperate reference (Figure 2). To separate climate from data availability we draw real missingness patterns from GHCN-Bangladesh windows and impose them, right-aligned and deterministically, on temperate contexts. Masking moves the temperate observed share to 0.808 for rainfall (Bangladesh reference 0.819) and 0.863 for mean temperature (reference 0.863). Crucially the mask is applied to the trained baselines as well as to the foundation models, so the control is like-for-like; masking only the model understates the remaining effect.

**Leakage audit.** Every model’s pretraining corpus was checked against its technical report rather than its model card. No corpus names GHCN-Daily, BMD or any Bangladesh source. The station corpora that do appear, USHCN and SubseasonalClimateUSA, are United States only, and our temperate reference is entirely European, so neither side of the tropical-versus-temperate contrast is contaminated. Leakage-free zero-shot evaluation is not itself novel [28]; what matters here is that it is a precondition for reading a geographic contrast as a geographic effect.

**Context length.** Because a sparse context is also a shorter effective context, the tropical contrast is repeated with 2,048 days of history in place of 1,024, with the control masked identically.

## 5 Results

We evaluate 8 zero-shot foundation models against 10 trained and statistical baselines on identical windows: Chronos-2 [4] and Chronos-Bolt [3], TimesFM 2.5 [7], Toto-2.0 [6], TiRex [5], Moirai-2 [19, 33], Granite TTM r2 [8] and Sundial [20]. All models see a 1,024-day context except TTM, whose daily variants cap at 512.

### 5.1 Foundation models lead, but most margins are not separable

Averaged over the 16 daily tasks, six of the eight foundation models outrank every trained baseline. Chronos-2 has the best mean rank at leads 7 and 30 (2.77 of 18 models overall), TiRex the best at lead 1, and the strongest trained baseline, PatchTST, ranks 6.96 (Table 1). Friedman rejects equality decisively at every lead ( $\chi^2 = 220.1, 215.5, 209.8; p < 10^{-34}$ ).

That ranking, however, does not license an ordering. The Nemenyi critical difference is 6.58, and the six leading foundation models together with PatchTST fall inside one critical difference of the best at leads 7 and 30 (Figure 4). The group separates clearly from the weaker baselines; positions within it do not.

Pairwise, the picture is more sober still. Comparing the best foundation model with the best baseline for each task and lead, and correcting across all 48 comparisons, the foundation model is significantly better in 14, worse in 5, and statistically indistinguishable in 29. The 14 wins concentrate at a one-day lead and on the smooth reanalysis tracks: partitioned by track type, foundation models win 3 of 24 comparisons on station observations, all at lead 1, against 11 of 24 on reanalysis at the same coordinates. Of 432 model-task-lead skill intervals, 272 exclude climatology.

Table 1: Main leaderboard over the 16 daily tasks. Ranks are Friedman average ranks across the tasks at each lead (1 is best, 18 models); *Overall* is the mean of the three. CRPSS is mean skill against the same track’s day-of-year climatology at a 7-day lead, so a positive value beats climatology and zero is climatology, averaged over the 8 station-observation tasks and the 8 reanalysis tasks separately. The Nemenyi critical difference is 6.58, so rank gaps smaller than that are not separable. Parameter counts apply to the zero-shot models; the baselines are fitted per series. TTM r2 is point-only and is scored on point skill.

|                                          | Params | Average rank (1 = best) |        |         |         | Mean CRPSS, lead 7 |            |
|------------------------------------------|--------|-------------------------|--------|---------|---------|--------------------|------------|
| Model                                    | (M)    | Lead 1                  | Lead 7 | Lead 30 | Overall | Station            | Reanalysis |
| <i>Zero-shot foundation models</i>       |        |                         |        |         |         |                    |            |
| Chronos-2                                | 119.5  | 4.19                    | 2.12   | 2.00    | 2.77    | 0.035              | 0.128      |
| TiRex                                    | 35.3   | 1.62                    | 3.44   | 3.62    | 2.90    | 0.025              | 0.121      |
| TimesFM 2.5                              | 231.3  | 4.38                    | 4.44   | 3.75    | 4.19    | 0.024              | 0.121      |
| Toto-2.0                                 | 312.7  | 3.75                    | 5.81   | 6.56    | 5.38    | −0.015             | 0.117      |
| Moirai-2                                 | 11.4   | 4.69                    | 6.31   | 7.06    | 6.02    | 0.003              | 0.110      |
| Chronos-Bolt                             | 205.3  | 5.81                    | 5.75   | 6.62    | 6.06    | 0.016              | 0.108      |
| Sundial                                  | 128.3  | 11.06                   | 10.69  | 10.31   | 10.69   | −0.088             | 0.031      |
| TTM r2                                   | 0.8    | 16.38                   | 15.94  | 15.25   | 15.85   | −0.374             | −0.246     |
| <i>Trained and statistical baselines</i> |        |                         |        |         |         |                    |            |
| PatchTST                                 | n/a    | 7.75                    | 6.31   | 6.81    | 6.96    | 0.009              | 0.080      |
| AutoARIMA                                | n/a    | 7.62                    | 8.62   | 8.88    | 8.38    | −0.062             | 0.025      |
| NHITS                                    | n/a    | 8.12                    | 8.94   | 8.88    | 8.65    | −0.002             | 0.023      |
| LightGBM                                 | n/a    | 12.88                   | 8.62   | 8.25    | 9.92    | −0.063             | 0.042      |
| DLinear                                  | n/a    | 13.25                   | 12.00  | 11.50   | 12.25   | −0.126             | −0.005     |
| AutoETS                                  | n/a    | 11.62                   | 14.44  | 15.81   | 13.96   | −0.255             | −0.180     |
| AutoTheta                                | n/a    | 12.94                   | 15.25  | 15.81   | 14.67   | −0.268             | −0.188     |
| <i>Naive references</i>                  |        |                         |        |         |         |                    |            |
| Climatology                              | n/a    | 14.62                   | 8.50   | 7.25    | 10.12   | 0.000              | 0.000      |
| Persistence                              | n/a    | 12.38                   | 16.69  | 17.38   | 15.48   | −0.444             | −0.278     |
| Seasonal naive                           | n/a    | 17.94                   | 17.12  | 15.25   | 16.77   | −0.406             | −0.348     |

## 5.2 Nothing beats climatology on tropical rainfall

All five baseline wins are a day-of-year climatology on rainfall at 7- and 30-day leads. On the 35-station BMD record no foundation model exceeds climatology beyond a one-day lead (best −0.004 CRPSS at lead 30), and on the independent CHIRPS dekadal task none beats it at any horizon: the best result, Chronos-2 given 28 years of context, is to match climatology to within 0.001 CRPSS (Table 2). The claim needs one scope marker: on the smaller 10-station GHCN-Bangladesh track the best foundation model does hold positive point estimates at every lead (Toto-2.0 +0.095 at day 1; Chronos-2 +0.019 at day 30), but none of those margins survives correction ( $p = 0.88, 1.00, 1.00$ ), so they are not separable from climatology either.

# F6 Critical-difference ranking: foundation models (blue) vs baselines (grey)

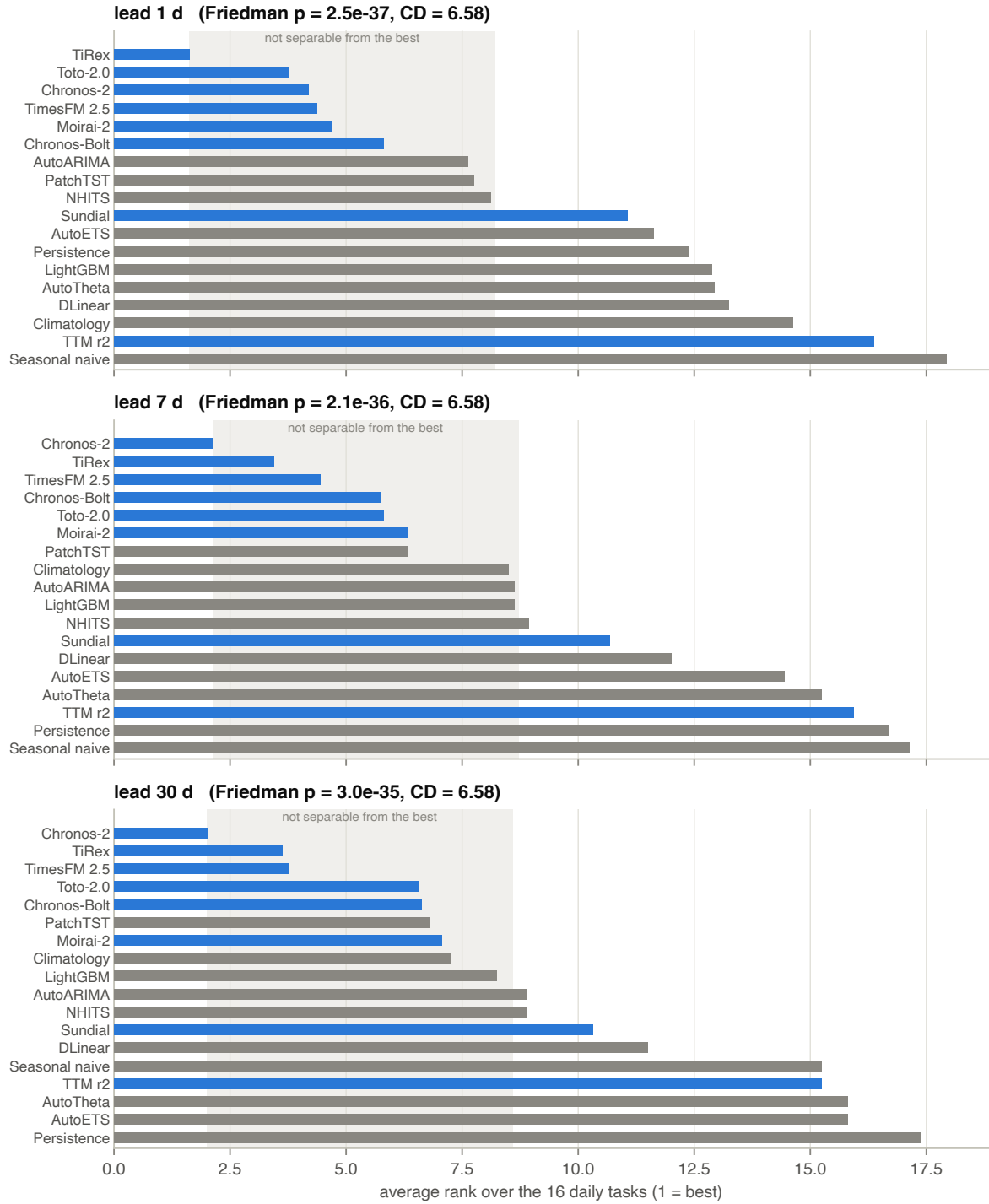


Figure 4: Average rank over the 16 daily tasks, with the shaded band marking one Nemenyi critical difference from the best model. Foundation models in blue, baselines in grey. The leading group is not internally separable.

Table 2: CHIRPS dekadal rainfall, skill against a dekad-of-year climatology (higher is better; 0 is climatology). Headline context is 144 dekads; the final column gives the best result with all available history.

| Model               | Lead 1 | Lead 2 | Lead 3 | Lead 6 | Best, 1,024 dekads |
|---------------------|--------|--------|--------|--------|--------------------|
| Climatology         | 0.000  | 0.000  | 0.000  | 0.000  | n/a                |
| Chronos-2           | -0.063 | -0.083 | -0.080 | -0.086 | -0.001             |
| Toto-2.0            | -0.086 | -0.115 | -0.114 | -0.113 | -0.004             |
| TimesFM 2.5         | -0.162 | -0.150 | -0.137 | -0.135 | -0.055             |
| TiRex               | -0.169 | -0.219 | -0.257 | -0.323 | -0.030             |
| Moirai-2            | -0.196 | -0.290 | -0.347 | -0.504 | -0.073             |
| Persistence         | -0.478 | -0.635 | -0.763 | -1.158 | n/a                |
| TTM r2 (point only) | -0.682 | -0.868 | -1.002 | -1.529 | -0.846             |

### 5.3 Reanalysis flatters models against the stations it represents

Scoring the same models on NASA POWER and at the co-located stations, skill is significantly higher on reanalysis in 78 of 108 model-variable-lead comparisons (Figure 5). The gap is largest for humidity (significant at every lead for all nine models tested) and rainfall (all nine at leads 1 and 30), smaller for temperature, and weakest for sunshine against radiation, two quantities that are not physically identical, which is why we do not lean on that row. The effect holds across model families, so it is a property of the target rather than of any architecture. A benchmark built on reanalysis therefore overstates both absolute skill and the margin foundation models hold over baselines: at a 7-day lead the best foundation model’s advantage over the best trained baseline is +0.026 on observations and +0.048 on reanalysis. This is the time-series counterpart of the critique Gupta et al. [14] make for AI weather-prediction models.

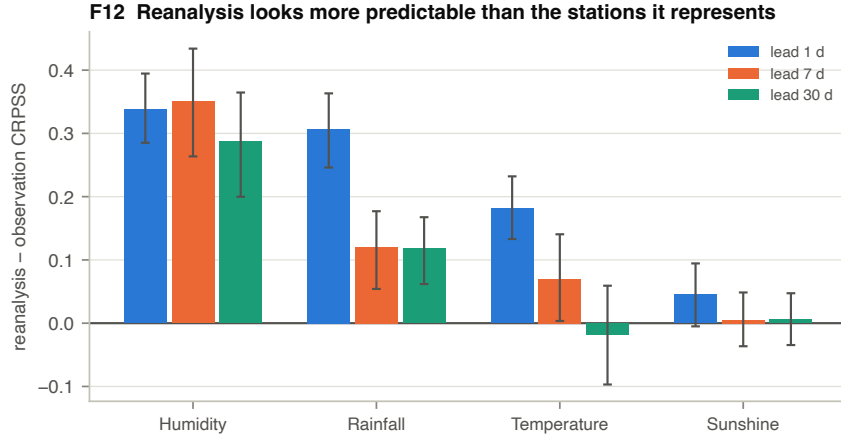


Figure 5: Skill on NASA POWER minus skill at co-located stations, with 95% moving-block intervals. Positive values mean the reanalysis looks more predictable than the observations it represents.

### 5.4 A long-lead tropical temperature deficit, mostly explained by sparse context

Against a trained-baseline control subjected to the same Bangladesh-like missingness, four of the eight foundation models retain a significant temperature penalty at a 30-day lead: TTM r2 (+0.187), Chronos-Bolt (+0.138), Chronos-2 (+0.060) and TiRex (+0.052). The median

penalty across the roster falls from +0.112 as observed to +0.043 once both sides face the same gaps, so roughly 60 % of the raw effect is attributable to sparser station context rather than to the tropics (Figure 6). At a one-day lead no model shows a significant penalty.

Two scope markers matter. First, masking only the model, as an ablation applied to the foundation models alone would, *understates* the effect, because the control also degrades on gappy contexts; the like-for-like comparison raises the median from +0.022 to +0.043 and returns TiRex to significance. Second, rainfall shows no systematic penalty at any lead: where results are significant at lead 1 the sign is reversed, with four models performing relatively *better* in Bangladesh than at temperate stations, and the single exception is Toto-2.0 at lead 7 (+0.028, CI 0.005 to 0.053).

Longer context does not dissolve what remains. Repeating the contrast with 2,048 days of history and the control masked identically leaves the median penalty at +0.052, unchanged from 1,024 days; Chronos-2 (+0.059) and TiRex (+0.052) stay significant, while TimesFM’s penalty, which was genuinely context-related, disappears. An unmasked comparison would have suggested the opposite, because without masking the median halves from +0.080 to +0.043 as context doubles. That apparent context effect is mostly the missingness effect in disguise.

**F8 Tropical temperature penalty, and how much of it is missing context**

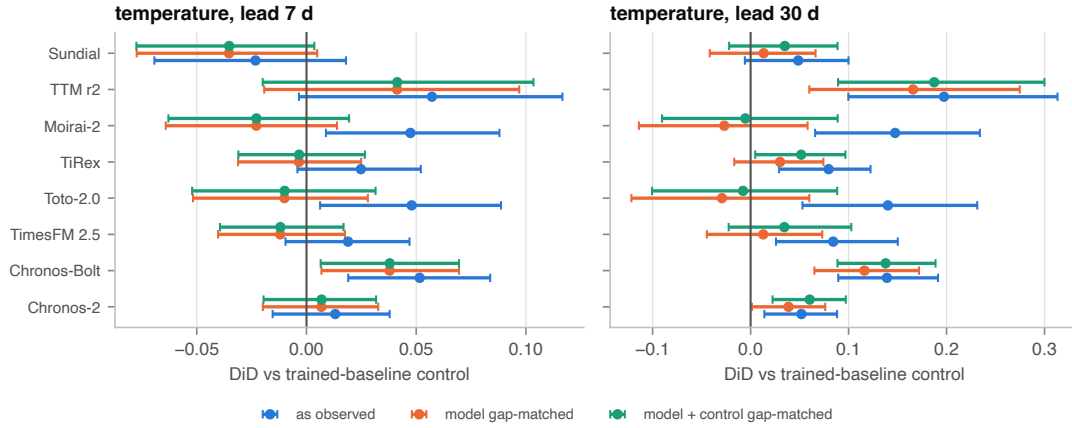


Figure 6: Difference-in-differences against the trained-baseline control, with 95% intervals. Positive values mean the model loses more skill in the tropics than the control does. Masking the control as well as the model is the like-for-like comparison.

## 5.5 Skill reverses by monsoon regime

Aggregate leaderboards hide the benchmark’s sharpest result. Stratifying July–August targets by spell, and blocking intervals by spell *event* rather than station-day, 61 of 72 contrasts are significant. In break spells every model is better than climatology, and the foundation models hold the widest margins: Chronos-Bolt reaches +0.440 on temperature at a 30-day lead against PatchTST’s +0.106. In active spells the ordering inverts: every model falls below climatology at 7- and 30-day leads, with the foundation models losing the most ground, Chronos-Bolt to −0.989 against PatchTST’s −0.193. Humidity repeats the pattern. Forecast quality is therefore lowest during the very episodes that drive flooding, transplanting and evacuation decisions.

## 5.6 Calibration and compute

Calibration varies more than accuracy. TimesFM 2.5 (0.831), Toto-2.0 (0.826) and TiRex (0.797) are close to the nominal 80 %; Chronos-2, the most accurate model overall, is sharp on rainfall (0.629); and Sundial is severely under-dispersed (0.323). A GPU diagnostic confirms the last

is a property of the model rather than of our adapter: interval width is flat in the number of sampled trajectories (1.44, 1.56, 1.59 for 20, 100 and 500 samples), the output shape matches the documented layout, and the trajectory spread of  $0.64^{\circ}\text{C}$  sits against the model’s own RMSE of  $1.51^{\circ}\text{C}$ .

The whole zero-shot leaderboard is cheap to reproduce. Every model fits in 4.2 GiB of GPU memory, and all 8 models over all 16 tasks cost 4.16 GPU-hours on a free Tesla T4, from 0.6 minutes for TTM r2 to 130 minutes for Sundial. By contrast AutoARIMA alone costs about 20.7 CPU-hours. Accuracy does not track size: TiRex ranks second overall with 35.3M parameters and 568 MiB, while Toto-2.0 (312.7M) ranks fourth and Sundial (128.3M) eleventh (Figure 7).

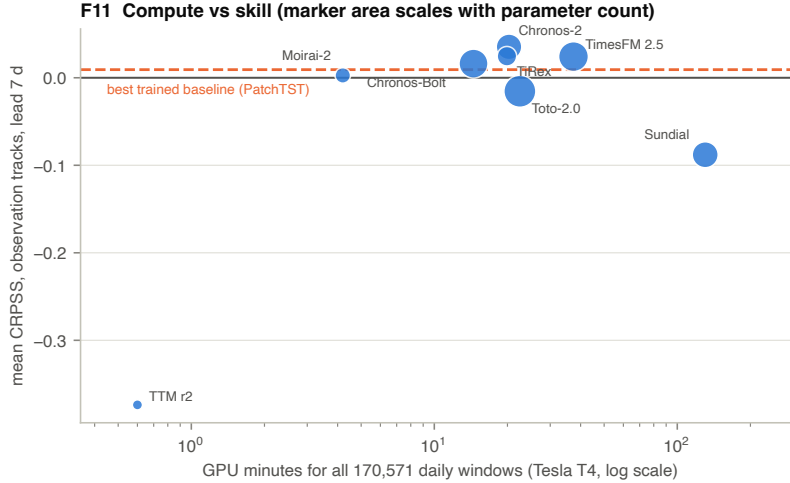


Figure 7: Compute against skill on the observation tracks at a 7-day lead; marker area scales with parameter count. The full leaderboard is reproducible in roughly four GPU-hours on free hardware.

## 6 Discussion

**What the nulls mean.** A reader could summarise this paper as “foundation models mostly tie”. That summary is accurate and is the point. The zero-shot claim is usually evidenced on aggregate leaderboards without significance testing; when the same models are scored on a tropical station network with the forecast origin as the unit of resampling and a correction across 48 comparisons, most margins on observations become indistinguishable from a well-built baseline. The models are not weak: six of eight outrank every trained baseline, and 272 of 432 skill intervals exclude climatology. But the size of the advantage on real station data is smaller than the literature’s framing implies.

**Why rainfall is different.** The rainfall result is not a modelling failure so much as a property of the target. Daily tropical rainfall is zero-inflated (67.6 % dry station-days) and convective, so beyond a day the predictable signal is largely the seasonal cycle, which a day-of-year climatology already encodes. A forecast that reproduces the climatological distribution is close to the achievable ceiling, and no model in our roster exceeded it beyond a one-day lead. This is worth stating plainly because rainfall is the variable on which the operational value of these models would be judged in Bangladesh.

**Why regime stratification matters.** The spell result is the clearest argument against aggregate reporting. The same models are strongly skilful during break spells and significantly

worse than climatology during active spells, a reversal that vanishes in a seasonal average. One mechanism would explain this, though we do not test it: a model with a long context window carries the persistent warm, dry conditions of a break forward, which is rewarded while the break holds and penalised as soon as convection resumes. Whatever the cause, a benchmark that reports only aggregate skill would certify these models as adequate for exactly the situation in which they are least reliable.

**Implications for benchmark design.** Three of our results are really statements about how evaluation is done. Scoring against reanalysis inflates apparent skill and, more subtly, inflates the *margin* between model classes, so reanalysis-only benchmarks are not a neutral convenience. Stratifying by a physically defined regime exposes failures that aggregates hide. And a context-availability control changes the interpretation of a geographic gap: without masking the trained baseline identically, our own earlier reading, that longer context dissolves the tropical deficit, did not survive.

**Affordability.** The entire zero-shot leaderboard reproduces in 4.16 GPU-hours on free hardware, and the strongest classical baseline we ran costs about five times that in CPU time. For the research communities this benchmark is about, zero-shot models are already the affordable option; the question this paper answers is what that affordability currently buys on their own data.

## 7 Limitations

**The equity contrast rests on ten stations.** The tropical-versus-temperate comparison uses the 10 GHCN-Daily stations in Bangladesh, giving 1,154 rainfall and 1,478 temperature windows, an order of magnitude fewer than the BMD track. Every equity figure in this paper should be read with that  $n$  attached. Three design choices limit but do not eliminate the risk: the temperate reference is drawn from the same instrument network, the gap-matched control is applied to models and baselines alike, and intervals come from a moving-block bootstrap over origins rather than from window counts. We report the contrast because it is the only like-for-like comparison available, not because ten stations settle the question.

**One country, one climate.** Findings are specific to Bangladesh’s monsoon regime. The protocol, controls and significance machinery transfer; the numbers do not. Whether the long-lead temperature deficit appears in other tropical networks is open.

**Zero-shot only.** We evaluate models as they are shipped. Few-shot adaptation and light fine-tuning, the obvious next step and cheap for several models in the roster, are out of scope, so this paper bounds the zero-shot claim rather than the models’ ceiling.

**The roster will age.** Eight models current in 2026 will not be current for long. The harness, cached forecasts and scoring code are released so that adding a model is a single run against unchanged windows.

**Imputation in the source record.** The published BMD data was imputed by its authors before release. We detect and flag suspect values and exclude flagged targets from scoring, but detection is imperfect; flagged shares over the test period run from 0.08 % (rainfall) to 4.44 % (humidity), with one station-variable at 14.1 %.

**Two comparisons are weaker than the rest.** Bright sunshine hours and downward short-wave radiation are not the same physical quantity, so that row of the observation-versus-reanalysis contrast is reported but not relied on. The CHIRPS dekadal task is an independent-source check at coarse resolution, not a headline.

**Mechanisms are hypothesised, not tested.** We measure the regime reversal and the context-availability effect; we do not identify the architectural cause of either.

## 8 Conclusion

We built a benchmark for daily weather forecasting on tropical-monsoon station observations, with the controls needed to interpret a tropical result: a same-network temperate reference, a reanalysis track at identical coordinates, and a gap-matched context control applied to models and baselines alike. Evaluating 8 zero-shot time-series foundation models against 10 trained and statistical baselines over 170,571 rolling-origin windows, under a significance protocol that treats the forecast origin as the unit of resampling, the picture is mixed: foundation models are the strongest family, but on station observations most of their advantage is not statistically separable, nothing beats a day-of-year climatology on tropical rainfall beyond a one-day lead, reanalysis flatters every model relative to the stations it represents, skill reverses between monsoon break and active spells, and the long-lead tropical temperature deficit that half the roster shows is mostly, though not entirely, a consequence of sparser station context rather than of the tropics.

For practitioners in data-sparse regions the practical reading is that zero-shot models are already affordable and already competitive on smooth variables, and that for rainfall a well-built climatology remains the thing to beat. For benchmark builders it is that scoring target, regime stratification and context availability are not implementation details: each one changed a conclusion in this study.

## Data and code availability

The benchmark definition, evaluation harness, cached forecasts, significance code and figure scripts are available at <https://github.com/ihpolash/BanglaWeatherBench>. Source data are redistributed only where licensing permits: the BMD station record is CC-BY-4.0 [35]; GHCN-Daily and NASA POWER are public domain; CHIRPS is open [10]. Derived tracks are rebuilt by the released pipeline rather than shipped. The release includes a datasheet in the sense of Gebru et al. [11], covering composition, collection, preprocessing, uses and limitations.

## Reproducibility

Model inference runs in per-model isolated environments; every forecast file is verified for row counts, window identity, lead coverage, missing values, quantile monotonicity and non-negativity before it enters a leaderboard. The headline numbers in this paper are recomputed from released artefacts by `scripts/audit_claims.py`.

## Declaration on the use of generative AI

The author used an AI assistant (Claude, Anthropic) for implementation support, analysis and figure scripting, and for drafting and editing this manuscript, in all cases under the author’s direction and review. The research questions, benchmark design, evaluation protocol and the interpretation of the results are the author’s. All numerical results were produced by the released

code and were recomputed from the released artefacts by an audit script before submission; every reference was verified by the author against its primary source. The author takes full responsibility for all content of this paper, irrespective of how it was drafted.

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